

Bai 1

```
Output(s):
Successfully stored 11724 records in: "/media/sf_DS200_BigData/Lab2/bai1_result"

Counters:
Total records written : 11724
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local377911738_0001 ->      job_local1870888967_0002,
job_local1870888967_0002

2026-04-08 23:18:37,010 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already
initialized!
2026-04-08 23:18:37,017 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already
initialized!
2026-04-08 23:18:37,028 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already
initialized!
2026-04-08 23:18:37,064 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already
initialized!
2026-04-08 23:18:37,068 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already
initialized!
2026-04-08 23:18:37,076 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already
initialized!
2026-04-08 23:18:37,085 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Succ
ess!
2026-04-08 23:18:37,147 [main] INFO  org.apache.pig.Main - Pig script completed in 24 seconds and 302 milliseconds (24302
ms)
quang@quang-VirtualBox: /media/sf_DS200_BigData/Lab2$
```

```
assignments.ipynb  bai1.pig  part-r-00000  hotel-review.csv
bai1_result > part-r-00000 > data
1 1;{(qua),(hotel),(vu),(san),(ttc),(lan),(dung),(vua),(lan),(khach),(dich),(ngoc),(ngoc),(da),(lat),(premium));general;hotel;positive
2 1;{(lan),(khach),(hotel),(ttc),(ngoc),(qua),(vu),(ngoc),(premium),(san),(lan),(lat),(vua),(da),(dich),(dung));general;service;positive
3 128;{(khach),(lan),(that),(san),(da),(qua),(hotel),(mat),(vong),(ttc),(ngoc),(ngoi),(ngoi),(lat),(premium),(lan),(chuyen),(vua),(ngoc),(ngoi));general;hotel;positive
4 129;{(ot),(et),(lon),(nhut),(san),(go),(duc),(khong),(thut),(kau),(ngu));comfort;rooms;negative
5 129;{(khong),(san),(thut),(nhut),(lon),(go),(kau),(duc),(et),(ot),(ngu));quality;room_amenities;negative
6 130;{(te),(do),(ky),(nhu),(chuyen),(khong),(u),(vien),(am),(spa),(nghiep),(phong),(thut),(cu),(nhan));general;service;negative
7 130;{(khong),(phong),(u),(chuyen),(am),(cu),(do),(vien),(spa),(nghiep),(ky),(nhan),(nhu),(te),(thut));quality;facilities;negative
8 131;{(so),(xung),(nao),(tuong),(chut),(tu),(tien),(khong),(thut),(gia));prices;hotel;neutral
9 132;{(lat),(tiou),(khong),(lan),(premium),(lan),(gia),(4),(khach),(xung),(san),(hotel),(sao),(ngoc),(ttc),(luong),(danh),(chuan),(ngoc),(chat),(da))
10 133;{(phong),(toi),(nhut),(thut),(dan));design&features;room_amenities;negative
11 134;{(am),(lan),(cu),(xuong),(do),(cap),(cach),(tot),(phong),(thut),(khong),(may),(khong),(dac));comfort;rooms;negative
12 134;{(do),(tot),(cap),(am),(xuong),(may),(thut),(phong),(khong),(dac),(cu),(cach),(khong),(lan));quality;rooms;negative
13 134;{(thut),(may),(cap),(xuong),(lan),(am),(dac),(cu),(khong),(khong),(cach),(tot),(do),(phong));quality;room_amenities;negative
14 134;{(cach),(phong),(dac),(khong),(do),(xuong),(may),(tot),(lan),(thut),(cap),(cu),(khong),(am));design&features;room_amenities;negative
15 135;{(xe),(san),(dau),(xe),(duong),(3),(khach),(ham),(chiec),(tu),(khong),(suot),(chi),(cho),(thut),(dau));design&features;facilities;negative
16 136;{(hotel),(lat),(ve),(oc),(sinh),(sach),(khach),(gang),(ngoc),(gon),(lan),(ngoc),(da),(phong),(ttc),(premium),(san),(lan));cleanliness;rooms;positive
17 136;{(lan),(san),(lat),(ngoc),(lan),(ngoc),(oc),(ve),(gon),(ttc),(sinh),(sach),(hotel),(phong),(da),(khach),(gang),(premium));design&features;rooms;positive
18 137;{(vien),(phuc),(than),(thai),(do),(nhan),(thien),(vu),(vuoi),(ve));general;service;positive
19 138;{(sinh),(dau),(that),(giu),(tien),(tri),(ngoi),(sach),(phong),(se),(bo),(thiet),(gin),(ve),(du),(noi));general;room_amenities;positive
20 138;{(tien),(sach),(gin),(ve),(se),(noi),(that),(phong),(dau),(ngoi),(du),(tri),(sinh),(thiet),(bo),(giu));cleanliness;rooms;positive
21 138;{(noi),(sach),(se),(ve),(sinh),(tri),(du),(ngoi),(dau),(bo),(tien),(giu),(phong),(gin),(thiet),(that));design&features;rooms;positive
22 139;{(dep),(xuong),(that),(phu),(khong),(tien),(khong),(trong),(hop),(chuan),(khong),(ngoi),(sang),(noi),(4),(cap),(giuong));quality;room_amenities;positive
23 140;{(sach),(dau),(cu),(du),(that),(noi),(ve),(sinh),(no),(san),(sach));general;room_amenities;positive
24 140;{(sach),(sinh),(san),(sach),(du),(no),(that),(dau),(noi),(ve),(cu));cleanliness;rooms;positive
25 141;{(phong),(cu),(mem),(an),(trai),(thuc),(tuy),(banh),(nhan),(lan),(tu));quality;facilities;negative
```

Bai 2

```
Input(s):
Successfully read 11724 records from: "/media/sf_DS200_BigData/Lab2/bai1_result/part-r-00000"

Output(s):
Successfully stored 8 records in: "/media/sf_DS200_BigData/Lab2/bai2_result/category_count"
Successfully stored 7 records in: "/media/sf_DS200_BigData/Lab2/bai2_result/aspect_count"
Successfully stored 34 records in: "/media/sf_DS200_BigData/Lab2/bai2_result/word_count"

Counters:
Total records written : 49
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1926390554_0001 ->      job_local465325690_0002,job_local48929978_0003,job_local1627499706_0004,
job_local465325690_0002
job_local48929978_0003
job_local1627499706_0004
```

```
2026-04-09 16:32:19,322 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 16:32:19,327 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 16:32:19,332 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 16:32:19,416 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 16:32:19,421 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 16:32:19,428 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 16:32:19,434 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-09 16:32:19,465 [main] INFO org.apache.pig.Main - Pig script completed in 19 seconds and 108 milliseconds (19108 ms)
```

```
quang@quang-VirtualBox: /media/sf_DS200_BigData/Lab2$
```

```
bai2_result > aspect_count > ≡ part-r-00000
```

```
1 hotel;3086
2 rooms;1791
3 service;2298
4 location;956
5 facilities;416
6 food&drinks;543
7 room_amenities;1156
8 |
```

```
bai2_result > category_count > ≡ part-r-00000
```

```
1 prices;740
2 comfort;1342
3 general;4667
4 quality;1091
5 cleanliness;1304
6 miscellaneous;328
7 style&options;265
8 design&features;882
9 |
```

```
assignments.ipynb  bai2.pig  _SUCCESS  part-r-00000  bai1_result  part-r-00000 ...word_count  X  ...
bai2_result > word_count >  part-r-00000 > data
1  cái;626
3  hai;731
4  nghi;775
5  như;549
6  sinh;558
7  trí;775
8  vệ;543
9  vị;744
10  vụ;1667
11  đà;771
12  ốc;831
13  hợp;522
14  lòng;742
15  lạt;658
16  nhân;1483
17  sạn;2124
18  tình;550
19  tốt;1036
20  viên;1469
21  đây;951
22  đủ;696
23  chất;622
24  dịch;996
25  khách;2580
26  không;1488
27  nghĩ;778
28  phòng;2385
29  phục;771
30  resort;752
```

Bài 3

```
Output(s):
Successfully stored 2 records in: "/media/sf_DS200_BigData/Lab2/bai3_result"

Counters:
Total records written : 2
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1148508109_0001 -> job_local1763741195_0002,
job_local1763741195_0002 -> job_local240529766_0003,job_local147320418_0004,
job_local240529766_0003 -> job_local1819368851_0005,
job_local1819368851_0005 -> job_local382025112_0008,
job_local382025112_0008 -> job_local119625113_0009,
job_local147320418_0004 -> job_local284234141_0006,
job_local284234141_0006 -> job_local712390844_0007,
job_local712390844_0007 -> job_local119625113_0009,
job_local119625113_0009 -> job_local54219588_0010,
job_local54219588_0010 -> job_local584806334_0011,
job_local584806334_0011

2026-04-09 17:19:45,034 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,040 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,046 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,076 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,081 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,085 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,093 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,095 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
```

```
2026-04-09 17:19:45,040 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,046 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,076 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,081 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,085 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,093 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,095 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,099 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,104 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,108 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,112 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,121 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,132 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,136 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,141 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,144 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,148 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,158 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,164 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,168 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,176 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,180 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,184 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,226 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,237 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,242 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:19:45,247 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-09 17:19:45,313 [main] INFO org.apache.pig.Main - Pig script completed in 21 seconds and 693 milliseconds (21693 ms)
quang@Quang-VirtualBox:/media/sf_DS200_BigData/Lab2$
```

```
bai3_result > part-r-00000
1 Positive Top,hotel,2483
2 Negative Top,rooms,409
3
```

Bài 4

```
job_local2066206672_0002 1 1 n/a n/a n/a n/a n/a n/a n/a final_neg,neg_category_group,sorted,top_5_neg 6
ROUP_BY

Input(s):
Successfully read 11724 records from: "/media/sf_DS200_BigData/Lab2/bai1_result/part-r-00000"

Output(s):
Successfully stored 80 records in: "/media/sf_DS200_BigData/Lab2/bai4_result"

Counters:
Total records written : 80
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local2049213020_0001 -> job_local1118729757_0003,job_local2066206672_0002,
job_local1118729757_0003 -> job_local1616166303_0004,
job_local2066206672_0002 -> job_local1616166303_0004,
job_local1616166303_0004

2026-04-09 17:30:50,807 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:30:50,817 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:30:50,825 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:30:50,923 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:30:50,937 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:30:50,944 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:30:50,967 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-09 17:30:51,032 [main] INFO org.apache.pig.Main - Pig script completed in 14 seconds and 978 milliseconds (14978 ms)
quang@Quang-VirtualBox:/media/sf_DS200_BigData/Lab2$
```

```
bai4_result > part-m-00000 > data
1 Positive,prices,giá,602
2 Positive,prices,cà,411
3 Positive,prices,hộp,320
4 Positive,prices,vụ,312
5 Positive,prices,phòng,271
6 Positive,comfort,lòng,541
7 Positive,comfort,hải,530
8 Positive,comfort,khách,440
9 Positive,comfort,phòng,436
10 Positive,comfort,sạn,382
11 Positive,general,khách,2426
12 Positive,general,phòng,2249
13 Positive,general,vụ,2021
14 Positive,general,nhân,1848
15 Positive,general,sạn,1827
16 Positive,general,giá,222
```

```
bai4_result > part-m-00001 > data
1 Negative,prices,giá,56
2 Negative,prices,không,47
3 Negative,prices,phòng,30
4 Negative,prices,tiền,23
5 Negative,prices,khách,21
6 Negative,comfort,phòng,215
7 Negative,comfort,không,197
8 Negative,comfort,khách,107
9 Negative,comfort,sạn,81
10 Negative,comfort,hải,70
11 Negative,general,không,371
12 Negative,general,phòng,337
13 Negative,general,khách,274
14 Negative,general,sạn,199
15 Negative,general,nhân,149
16 Negative,quality,không,276
```

Bài 5

```
job_local556207011_0002 1      1      n/a      n/a      n/a      n/a      n/a      n/a      n/a      n/a      grouped_by_category,sorted,top_5_related  GROUP_BY
job_local1933078753_0003 1      1      n/a      n/a      n/a      n/a      n/a      n/a      n/a      n/a      final_result      SAMPLER

Input(s):
Successfully read 11724 records from: "/media/sf_DS200_BigData/Lab2/bai1_result/part-r-00000"

Output(s):
Successfully stored 40 records in: "/media/sf_DS200_BigData/Lab2/bai5_result"

Counters:
Total records written : 40
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1659040951_0001      ->      job_local556207011_0002,
job_local556207011_0002 ->      job_local1933078753_0003,
job_local1933078753_0003 ->      job_local266891667_0004,
job_local266891667_0004

2026-04-09 17:47:37,044 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:47:37,048 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:47:37,053 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:47:37,120 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:47:37,127 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:47:37,129 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-09 17:47:37,138 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-09 17:47:37,190 [main] INFO org.apache.pig.Main - Pig script completed in 16 seconds and 519 milliseconds (16519 ms)
quang@Quang-VirtualBox:/media/sf_DS200_BigData/Lab2$
```

```
bai5_result > part-r-00000 > data
1 style&options,không,101
2 style&options,bữa,115
3 style&options,món,150
4 style&options,ăn,188
5 style&options,sáng,225
6 quality,không,406
7 quality,phòng,608
8 quality,khách,454
9 quality,chất,452
10 quality,lượng,417
11 prices,giá,697
12 prices,phòng,320
13 prices,hợp,334
14 prices,vụ,338
15 prices,cà,451
16 miscellaneous,nước,84
17 miscellaneous,phòng,170
18 miscellaneous,khách,145
19 miscellaneous,sạn,90
20 miscellaneous,không,207
21 general,vụ,2293
22 general,nhân,2100
23 general,sạn,2185
24 general,phòng,2749
25 general,khách,2882
26 design&features,khách,303
27 design&features,không,400
28 design&features,đẹp,302
29 design&features,phòng,786
```